

ANVIL Hackathon – Autonomous Multi-Agent Repository Intelligence System

1. Agent Architecture

The system follows a multi-agent architecture where specialized AI agents collaborate through a centralized orchestrator.

Agents

- **Repository Analysis Agent**
 - Understands repository structure, frameworks, and modules.
- **Pull Request Analysis Agent**
 - Analyses code diffs and detects risky changes.
- **Security & Quality Agent**
 - Identifies vulnerabilities, unsafe patterns, and maintainability issues.
- **Reflection Agent**
 - Validates findings and removes duplicate or low-confidence outputs.
- **Summary Agent**
 - Aggregates results and generates stakeholder-specific summaries.

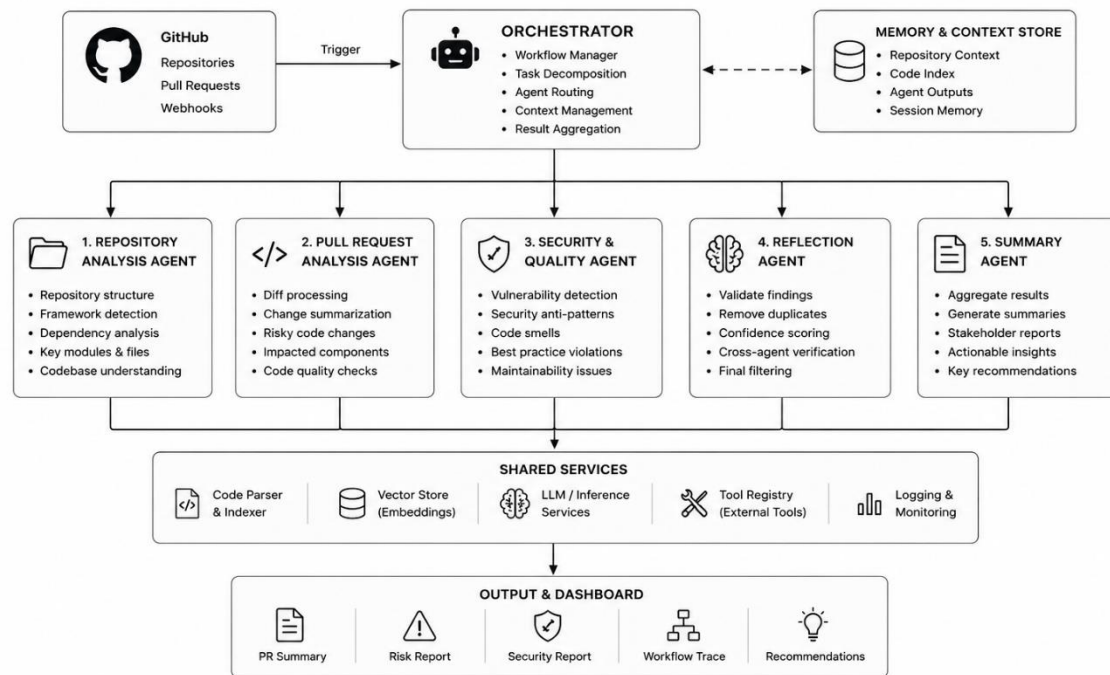
Orchestrator

The orchestrator manages:

- Agent execution
- Workflow sequencing
- Shared context
- Result aggregation

AGENT ARCHITECTURE

Multi-Agent AI System for Repository & Pull Request Analysis



2. Business Use Case

Modern development teams handle large repositories and frequent pull requests, making manual review slow and difficult.

Our system acts as an autonomous AI-powered engineering assistant that:

- Analyses repositories
- Reviews pull requests
- Detects risks and vulnerabilities
- Generates summaries automatically

Primary Users

- **Developers** → Receive bug warnings and PR summaries.
- **Code Reviewers** → Get highlighted risky changes and security alerts.
- **Project Managers** → View workflow summaries and repository insights.

Business Value

- Faster pull request reviews
- Reduced manual effort
- Improved code quality
- Early risk detection

- Better collaboration
-

3. Overall Workflow

1. A pull request is created or updated on GitHub.
2. The orchestrator triggers the multi-agent pipeline.
3. Repository agents analyse project structure and dependencies.
4. PR analysis agents inspect code changes and diffs.
5. Security agents detect vulnerabilities and risky patterns.
6. Reflection agents validate findings.
7. The summary agent generates final reports.
8. Results are displayed on the dashboard for stakeholders.